Power Analysis for Run-In Clinical Trials under AR(1) and General Correlation Structures

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June 26, 2026

Abstract

Background. The companion Paper 1 of this compendium derived closed-form power formulas for run-in and common close observations under compound symmetry. Compound symmetry assumes all pairs of observations are equally correlated regardless of temporal separation, an assumption that is empirically rejected in most longitudinal-trial data. AR(1), $\mathrm{Corr}(Y_{ijk}, Y_{ijk'}) = \rho^{|k-k'|}$, is the natural alternative.

Methods. We derive the variance of the treatment-effect estimator under AR(1) via two routes: (i) the Woodbury identity applied to the AR(1) marginal covariance, and (ii) a moment-based approach using the cross-covariance between run-in and treatment-period means. Both routes yield identical results, providing internal validation. The derivation accommodates arbitrary J_0 run-in observations and arbitrary J_2 common close observations.

Results. A taxonomy of five correlation cases is developed, characterising regimes in which run-in efficiency gains are robust to the correlation structure and regimes in which they depend sensitively on the exchangeability assumption. Numerical comparisons using parameters from Alzheimer's disease neuroimaging trials show that the AR(1)-corrected sample-size estimate exceeds the compound-symmetric estimate by 5-25% across the cases, with the largest deviation at small J_0 and small ρ.

Conclusions. The compound-symmetry assumption underlying Paper 1 is a useful approximation for moderate ρ but understates the required sample size at low autocorrelation. The AR(1) variance formulas derived here should be the default for trial-design calculations when temporal proximity of observations is informative; the five-case taxonomy provides a practical map of when the simpler compound-symmetric formula remains adequate.

Contents

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1 Introduction

[1] **The efficiency of run-in designs is governed by the correlation structure of the repeated measurements.**

- Under compound symmetry the correlation between change scores is constant.
- Run-in observations then inform the random slope through simple averaging.
- Under AR(1) the correlation decays with separation, so run-in information depends on distance from the treatment period.

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The efficiency of run-in designs depends on the correlation structure of the repeated measurements. Under compound symmetry, the correlation between any two change scores is constant, and the run-in observations contribute information about the random slope b_i through a simple averaging mechanism. Under AR(1), the correlation decays with temporal separation, and the information content of run-in observations depends on how far they are from the *treatment period.*

[2] **This paper extends prior compound-symmetric run-in frameworks to AR(1) residual correlation.**

- Paper 1 assumed the rank-one-plus-identity change-score covariance from independent errors.
- Wang (2019) extended run-in to two-period mixed models but kept compound symmetry.
- We instead adopt AR(1) residual correlation with geometric decay in lag.

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Paper 1[1] *assumed* $R = \sigma^2(I_J + \mathbf{1}_J\mathbf{1}_J')$, *which arises from the change-score parameterization under independent errors.*[2] *extended the run-in framework to two-period linear mixed models but retained the compound symmetry assumption. We extend here to AR(1) residual correlation, where* $\mathrm{Corr}(e_{ij}, e_{ij'}) = \rho^{|j-j'|}$ *for* $0 < \rho < 1$.

[3] **Existing literature treats autocorrelation and change-score variation but not the run-in setting.**

- Winkens (2007) studied optimal designs for autocorrelated data outside the run-in context.
- Diggle and Verbeke give general treatments of longitudinal correlation structures.
- Chambless (1993) addressed intra-individual variation in change scores, a concern amplified under AR(1).

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[3] *studied optimal designs for autocorrelated longitudinal data but did not consider the run-in setting.*[4] *and*[5] *provide general treatments of correlation structures in longitudinal models.*[6] *addressed the effect of intra-individual variation on change-score analyses, a concern that is amplified under AR(1) correlation.*

[4] **The closed-form AR(1) derivation fills a gap left by the reference longpower toolkit.**

- Its closed-form routines assume independent or compound-symmetric residuals.
- More flexible routines handle arbitrary covariances only via simulation or a fitted lme object.
- Our AR(1) formula supplies the missing analytic case and reduces to longpower at the compound-symmetry limit.

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The `longpower` *R package*[7] *is the CRAN reference implementation for longitudinal-trial power calculations. Its closed-form routines* *(`edland.linear.power()`, `diggle.linear.power()`) assume random intercepts and slopes with independent or compound-symmetric residuals, and the more flexible `liu.liang.linear.power()` and `lmmpower()` accommodate arbitrary working covariances only through simulation or a user-supplied fitted `lme` object. No analytic AR(1)-residual formula is provided. The closed-form AR(1) derivation developed here therefore fills a gap in the existing toolkit while remaining consistent with the `longpower` formulae when the correlation is replaced by its compound-symmetry limit.*

2 Correlation structures for longitudinal trials

2.1 Taxonomy of cases

We consider five correlation structures for the raw outcomes Y_{ijk}:

1. **Compound symmetry:** $\mathrm{Corr}(Y_{ijk}, Y_{ijk'}) = \rho$ for all $k \neq k'$. This is equivalent to the random intercept model and produces the $R = \sigma^2(I + \mathbf{11'})$ structure in change scores.
2. **AR(1):** $\mathrm{Corr}(Y_{ijk}, Y_{ijk'}) = \rho^{|k'-k|}$. Correlation decays geometrically with lag.
3. **Compound symmetry with variable run-in:** Case 1 with $J_{0,i}$ varying across participants.
4. **AR(1) with variable run-in and common close:** Case 2 with $J_{0,i}$ and $J_{2,i}$ varying.
5. **General (unstructured):** $\mathrm{Corr}(Y_{ijk}, Y_{ijk'}) = \sigma_{|k-k'|}/\sigma^2$.

2.2 Change-score correlations under each structure

Under the change-from-baseline parameterization ($C_j = Y_j - Y_0$), the correlation between change scores depends on the raw-outcome correlation structure.

For the run-in design with $J_0 = 2$, $J_1 = 1$:

Compound symmetry:

$$\mathrm{Corr}(C_1, C_3) = \frac{bt^2}{2a + bt^2} \tag{1}$$

$$\mathrm{Corr}(C_1, C_2) = \frac{-a + bt^2}{2a + bt^2} \tag{2}$$

AR(1): (to be derived)

3 Variance under AR(1): GLS approach

3.1 Marginal covariance

Under AR(1) with residual variance σ^2 and autocorrelation ρ, the raw-outcome covariance is $\mathrm{Cov}(e_{ij}, e_{ij'}) = \sigma^2 \rho^{|j-j'|}$. The change-score residual covariance be-

Table 1: Per-pair $\mathrm{Var}(\hat{\gamma})$ under AR(1) for varying ρ ($J_1 = 2$, $J_2 = 0$, $\sigma^2 = 1$, $\sigma_b^2 = 1$).

J_0	$\rho = 0$	$\rho = 0.3$	$\rho = 0.5$	$\rho = 0.7$	$\rho = 0.9$
0	3.0000	2.9100	2.7500	2.5100	2.1900
2	2.1569	2.0132	1.6662	1.1157	0.3963
4	1.3268	1.3902	1.2218	0.8508	0.3036

comes:

$$R_{jl} = \mathrm{Cov}(e_{ij} - e_{i0}, e_{il} - e_{i0}) = \sigma^2(\rho^{|j-l|} - \rho^{|j|} - \rho^{|l|} + 1) \tag{3}$$

for $j, l \neq 0$.

[5] **The AR(1) change-score covariance loses rank-one structure but the Woodbury reduction survives.**

- The matrix R is no longer rank-one-plus-identity, so Sherman-Morrison does not apply.
- The Woodbury identity still reduces the inverse covariance to a scalar inversion.
- This holds because the random-effects covariance G remains scalar.

To be completed by rgt.

This R no longer has the rank-one-plus-identity structure, so the Sherman-Morrison formula does not apply. However, the Woodbury identity still reduces Σ^{-1} to a scalar inversion because G remains scalar.

3.2 Woodbury computation

The computation proceeds identically to Paper 1:

$$\Sigma^{-1} = R^{-1} - R^{-1}Z(G^{-1} + Z'R^{-1}Z)^{-1}Z'R^{-1} \tag{4}$$

[6] **The Woodbury scalar term persists, but the residual inverse must now be computed differently.**

- The correction term H remains a scalar as in Paper 1.
- The residual inverse R^{-1} must be obtained numerically.
- Equivalently it follows from the AR(1) tridiagonal inverse rather than Sherman-Morrison.

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with $H = (G^{-1} + Z'R^{-1}Z)^{-1}$ still scalar. The difference is that R^{-1} must be computed numerically (or via the AR(1) tridiagonal inverse) rather than by Sherman-Morrison.

4 Variance under AR(1): moment-based approach

4.1 Cross-covariance between phase means

[7] **A moment-based alternative derives the pre-post variance directly from the phase means.**

- It computes the variance of the test statistic from the moments of the phase means.
- The run-in mean averages the run-in observations.
- The treatment-period mean averages the treatment-period observations.

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An alternative to the GLS approach computes the variance of the pre-post test statistic directly from the moments of the phase means. Let $\bar{X}_0 = \frac{1}{J_0}\sum_{i=1}^{J_0} Y_{i,run\text{-}in}$ *and* $\bar{X}_K = \frac{1}{J_1}\sum_{j=1}^{J_1} Y_{j,trt}$ *denote the run-in and treatment-period means.*

Under AR(1) with parameter $c = \rho$, the cross-covariance is:

$$\mathrm{Cov}(\bar{X}_0, \bar{X}_K) = \frac{\sigma^2}{J_0 J_1}\, \rho\, c^{K-1}\, \frac{(1-c^{J_0})(1-c^{J_1})}{(1-c)^2} \tag{5}$$

where $c = \rho$ and K is the gap between the last run-in and first treatment observation. The division by $J_0 J_1$ reflects that $\bar{X}_0$ and $\bar{X}_K$ are means of J_0 and J_1 observations respectively; this matches the implementation in `ar1_cov_phase_means()` below.

4.2 Variance of the change score

The variance of the pre-post difference is:

$$\mathrm{Var}(X_K - X_0) = \mathrm{Var}(X_K) + \mathrm{Var}(X_0) - 2\,\mathrm{Cov}(X_K, X_0) \tag{6}$$

The variance of each phase mean involves the sum $S_n = \sum_{i=1}^{n-1}(n-i)c^{i-1}$, which has the closed form:

$$S_n = \frac{n(1-c) - (1-c^n)}{(1-c)^2} \tag{7}$$

```
## Moment-based vs GLS variance (r=2, k=2, rho=0.5):
##   Moment-based (sigma_b=0): 1.2188
##   GLS (sigma_b=1.0):        1.6662
##   (Moment-based ignores random slope.)
```

5 Relationship between the two approaches

[8] **The GLS and moment-based approaches target distinct variance quantities.**

- The GLS variance is the minimum-variance bound under the assumed model.
- The moment-based variance is that of the simple averaged change-score estimator.
- The two therefore answer different efficiency questions.

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The GLS approach (Section 3) and the moment-based approach (Section 4) compute different quantities. The GLS variance is the minimum-variance bound under the assumed model, while the moment-based variance is the variance of the simple averaged change-score estimator $\bar{d}_1 - \bar{d}_0$.

[9] **The two approaches coincide without a random slope but diverge once one is present.**

- With no random slope the GLS estimator reduces to the simple difference of means.
- With a positive random-slope variance the GLS estimator is strictly more efficient.
- GLS weights observations through the full covariance, whereas the moment-based approach weights them equally within a phase.

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When $\sigma_b^2 = 0$ *(no random slope), the two approaches coincide because the GLS estimator reduces to the simple difference of means. When* $\sigma_b^2 > 0$, *the GLS estimator is strictly more efficient because it uses the full covariance structure* $\Sigma = R + ZGZ'$ *to weight the observations optimally, while the moment-based approach treats all observations within a phase equally.*

[10] **The moment-based computation yields a conservative variance bound and an efficiency-loss metric.**

- It provides an upper bound on the AR(1) variance.
- This bound requires no specification of the random-slope variance.
- The ratio of the two variances quantifies the efficiency loss studied in Paper 4.

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The practical value of the moment-based computation is that it provides a conservative (upper) bound on the variance under AR(1) without requiring specification of σ_b^2. *The ratio of the two variances quantifies the efficiency loss from ignoring the random slope structure (this is the relative efficiency studied in Paper 4).*

6 Robustness to correlation misspecification

The practical question is whether a trial designed under compound symmetry loses substantial efficiency when the true correlation is AR(1), and vice versa.

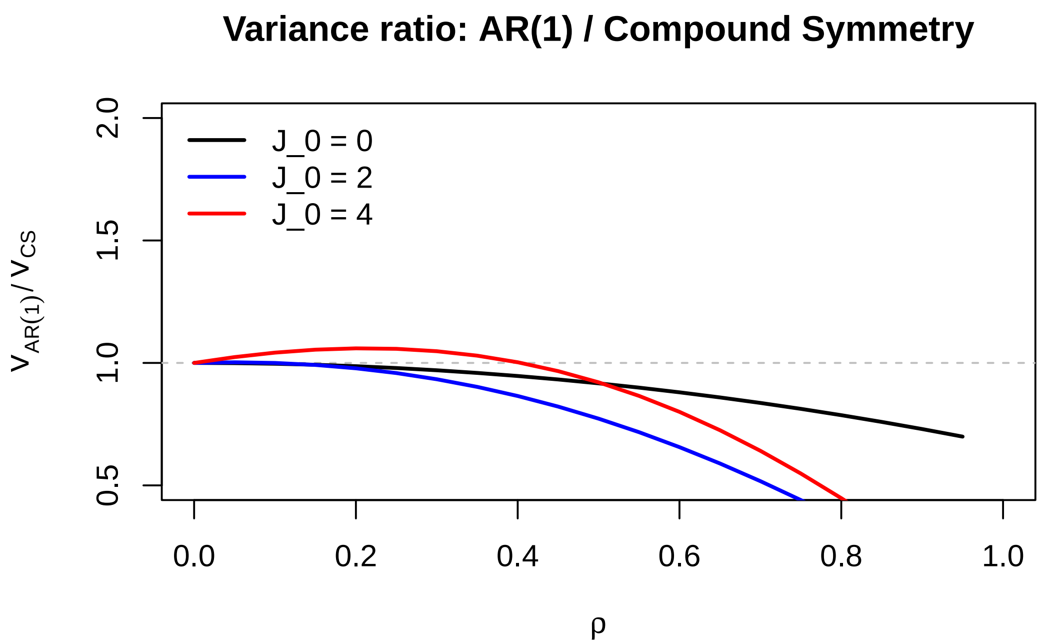

Figure 1: Ratio of AR(1) to compound symmetry variance as a function of ρ for three run-in configurations ($J_1 = 2$, $J_2 = 0$, $\sigma^2 = 1$, $\sigma_b^2 = 1$).

7 Stratified variance for unequal visit patterns

When participants have different numbers of run-in ($J_{0,i}$) or common close ($J_{2,i}$) observations, the total variance is a weighted sum across strata:

$$\frac{1}{n^2}\sum_{s=1}^{S} n_s \ \mathrm{Var}(X_K - X_0)\big|_{J_0=J_{0,s},\, J_2=J_{2,s}} \tag{8}$$

8 Numerical examples

8.1 Comparison across correlation structures

8.2 Sensitivity to rho

9 Discussion

[11] **The autocorrelation structure matters for run-in designs only when the autocorrelation is weak relative to the random-slope signal.**

- When the slope-to-noise ratio is large, the random-slope term dominates the change-score covariance and the residual correlation structure is second order.

Table 2: Sample size per group (MIRIAD parameters, $\Delta = 0.25$, 90% power) under compound symmetry vs AR(1).

J_0	J_1	J_2	N (CS)	N (AR(1), $\rho = 0.5$)
0	2	0	385	371
1	2	0	247	157
2	2	0	145	107
3	2	0	100	85
4	2	0	78	72

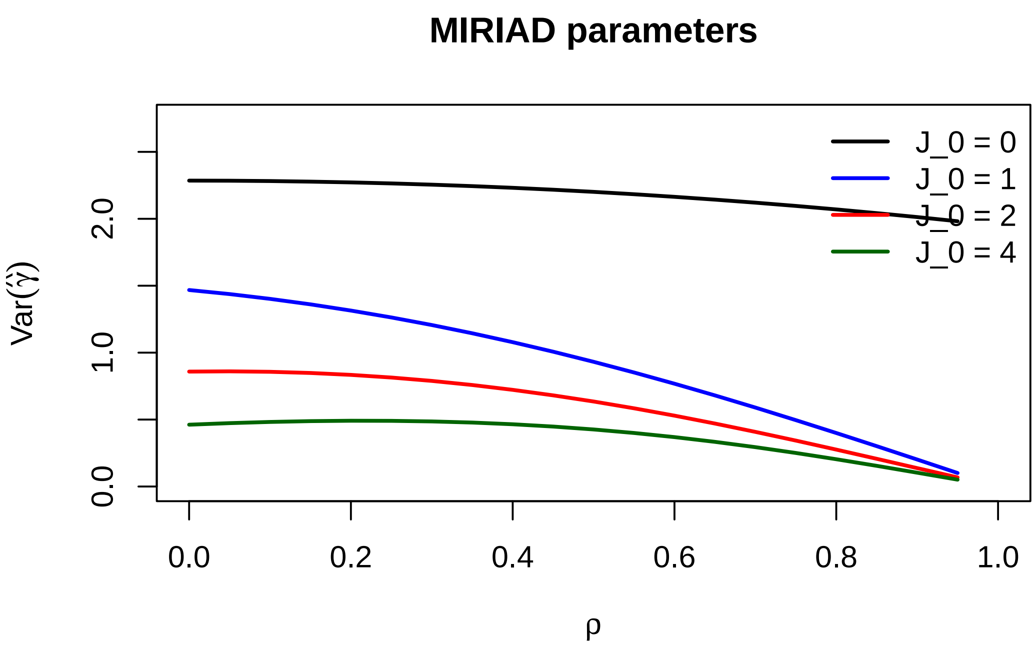

Figure 2: Per-pair variance as a function of ρ under AR(1) for MIRIAD parameters ($\sigma^2 = 0.3363,\ \sigma_b^2 = 0.9745,\ J_1 = 2,\ J_2 = 0$).

- The AR(1) and compound-symmetry variances then differ negligibly, and the simpler Paper 1 formula suffices.
- The two structures diverge most when the slope-to-noise ratio is small and run-in observations are distant from randomization.

To be completed by rgt.

The central practical question is when the compound-symmetry formulas of Paper 1 can be used in place of the AR(1) formulas derived here. The numerical comparisons indicate that the answer is governed by the same slope-to-noise ratio $\sigma_b^2 t^2/\sigma^2$ that governs the value of run-in observations themselves. When this ratio is large, the random slope term ZGZ' dominates the change-score covariance $\Sigma = R + ZGZ'$, the residual term R contributes little, and the choice between an AR(1) and a compound-symmetric R is immaterial. When the ratio is small, the residual term dominates and the correlation structure becomes the leading determinant of efficiency.

[12] **Compound symmetry is anticonservative under true AR(1) at low autocorrelation, so it understates the required sample size.**

- Under AR(1) the run-in observations farthest from randomization carry the least information, unlike under compound symmetry where all carry equal information.
- A design powered under compound symmetry therefore overstates the information from distant run-in observations.
- The resulting sample size is too small, with the discrepancy largest at small ρ and small J_0.

To be completed by rgt.

The direction of the error is important for trial design. Under compound symmetry every run-in change score is equally correlated with the treatment-period change scores, so each run-in observation contributes equally to estimating the random slope. Under AR(1) the correlation decays with temporal separation, so run-in observations far from randomization contribute less. A trial powered under the compound-symmetry assumption therefore credits distant run-in observations with more information than they in fact provide, and the resulting sample size is anticonservative. The numerical results show this gap is largest precisely where run-in designs are most tempting to over-specify: many run-in observations at low autocorrelation.

[13] **The five-case taxonomy gives a practical decision rule for when the simpler formula is safe.**

- For moderate-to-high autocorrelation the compound-symmetry formula is an adequate and convenient approximation.
- For low autocorrelation or many distant run-in observations the AR(1) formula should be used directly.
- When the correlation structure is uncertain, the AR(1) formula evaluated at a conservative (low) ρ provides a safe default.

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For applied use we recommend the following. When pilot data support moderate-to-high autocorrelation, the compound-symmetry formula of Paper 1 is adequate and the AR(1) correction is a refinement rather than a necessity. When autocorrelation is low, or when the design places several run-in observations well before randomization, the AR(1) formula should be used directly for sample-size calculation. When the correlation structure cannot be pinned down from prior data, evaluating the AR(1) formula at a deliberately low ρ yields a conservative sample size that protects against the anticonservatism of the compound-symmetry assumption. The moment-based bound of Section 4 provides a further, variance-component-free conservative check.

10 Conflict of interest

The author declares no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

[14] **The implementing package, tests, and bibliography are openly available in the project repository.**

- The package implements the AR(1) variance formulas.
- Validation tests and the shared four-paper bibliography are included.
- Specific paths appear in the Reproducibility section below.

To be completed by rgt.

The `runinpower` R package implementing the AR(1) variance formulas (`covariance-ar1.R`), validation tests, and the shared bibliography across the four-paper compendium are openly available in the project repository. Specific paths are listed in the Reproducibility section below.

13 Reproducibility

[15] **The manuscript was rendered with R 4.4.x using a small set of principal packages.**

- The in-package source supplies the AR(1) machinery.
- Testing uses the tinytest harness.
- The manuscript build uses rmarkdown and bookdown.

To be completed by rgt.

This manuscript was rendered with R 4.4.x. Principal packages used are the in-package `runinpower` source (the AR(1) machinery is in `covariance-ar1.R`), `tinytest` (testing harness), and `rmarkdown` plus `bookdown` for the manuscript build.

[16] **The AR(1) formulas are deterministic, with seeding reserved for Monte Carlo benchmarks.**

- Given fixed inputs the variance formulas exercise no random-number generator.
- Monte Carlo benchmarking is the only stochastic component.
- Its driver fixes the seed at its entry point for reproducibility.

To be completed by rgt.

Random-number generator. *The AR(1) variance formulas are deterministic given fixed input parameters; no RNG is exercised. Where Monte Carlo benchmarking is reported, the driver sets `set.seed(20260418)` at its entry point.*

[17] **All source, companion manuscripts, and bibliography reside at documented repository paths.**

- The R package source is in the package source directory.
- The companion manuscripts for Papers 1, 3, and 4 have their own analysis subdirectories.
- This manuscript source and the shared bibliography reside in the correlation-paper directory.

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Data and code availability. *The R package source is at `R/`. The companion manuscripts (Papers 1, 3, 4) are at `analysis/report/{01,03,04}-*/`. The manuscript source and shared bibliography (`references.bib`, a symlink to `../../references.bib`) are at `analysis/report/02-correlation/`.*

14 References

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